

Extending Automatic Prompt Optimization to Vision-Language Tasks

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GYYP

Abstract

Promptmatix automates prompt optimization for text-based Large Language Models but currently lacks multimodal capabilities (Murthy et al., 2025). This project extends the framework to support Vision-Language Models (VLMs) by implementing a highly technical Visual Question Answering (VQA) pipeline. We propose architectural modifications to programmatically ingest VQA data, inject static image contexts into DSPy signatures, synthesize multimodal datasets via external APIs, and engineer an LLM-as-a-Judge evaluation metric. This contribution enables automated, cost-aware prompt optimization for complex vision-language reasoning tasks without relying on manual UI interactions.

1 Introduction

Large Language Models depend heavily on well-crafted prompts, a process Promptmatix automates by transforming natural language descriptions into optimized prompts without manual tuning. However, the framework's single-prompt batch optimization model currently struggles with complex interaction patterns involving images. The original authors explicitly identify supporting multimodal prompt types as a critical future direction (Murthy et al., 2025). Extending this system requires persistent state, context-aware reasoning, and robust programmatic evaluation. Our approach implements a tech-heavy VQA pipeline, focusing on algorithmic feedback, token economics, and robust API ingestion to seamlessly process and optimize multimodal prompts.

2 Approach

We will modify the core Promptmatix components (Murthy et al., 2025) to support VQA:

2.1 VQA Ingestion & API Engineering

We will rewrite the Data Configuration module to programmatically ingest local image files or

URLs, compress them, and convert them to base64 strings. The LLM Configuration will route tasks to VLMs utilizing a robust API connector for provider-specific multimodal JSON payloads.

2.2 DSPy Pipeline & Token Economics

We will inject an `image_context` field into dynamically generated DSPy signatures. Images will be locked as static variables during MIPROv2 optimization so the algorithm only mutates text instructions. We will also update the cost-aware objective ($\mathcal{L} = \mathcal{L}_{performance} + \lambda \cdot \mathcal{L}_{cost}$) to accurately calculate the distinct token cost of static images (Murthy et al., 2025).

2.3 Multimodal Data Synthesizer

To solve the data bottleneck, the Synthetic Data Generation module will connect to external VQA dataset APIs (e.g., HuggingFace datasets like VQAv2) to sample image-text-answer triplets, automatically bootstrapping the train and val sets based on the user's task type.

2.4 Algorithmic Feedback & Metric Engineering

We will implement an LLM-as-a-Judge metric within the Evaluation module to handle the nuances of VQA ground-truth matching. Leveraging the framework's automatic Feedback Generation Module, we will program a reasoning model to analyze failed optimization trials, diagnose visual detail omissions, and automatically feed programmatic critiques back into the MIPROv2 loop.

3 Project Timeline

We still haven't chosen which of us will concentrate on which part and it we will certainly all be active on all parts, but to simplify things, we will set four main tasks which will each be attributed to one of us.

3.1 Weeks 1-2 (Milestone 1)

Member 1 implements VQA ingestion, base64 conversion, and API payload formatting. Member 2 integrates the image_context field into DSPy signatures and writes the logic for accurate vision token economics calculation.

3.2 Weeks 3-4 (Milestone 2)

Member 3 builds the data synthesizer script connecting to external VQA datasets to automatically bootstrap training sets. Member 4 engineers the LLM-as-a-Judge metric and programs the automated feedback generation module.

3.3 Weeks 5-6 (Final Submission)

System integration, rigorous algorithmic testing against multimodal benchmarks, and final report compilation (All Members).

References

Rithesh Murthy, Ming Zhu, Liangwei Yang, Jielin Qiu, Juntao Tan, Shelby Heinecke, Silvio Savarese, Caiming Xiong, and Huan Wang. 2025. Promptomatix: An automatic prompt optimization framework for large language models. *arXiv preprint arXiv:2507.14241*.